

LAB 1 SOLUTION DOCUMENT

Course: Cloud Computing (CC-F26)

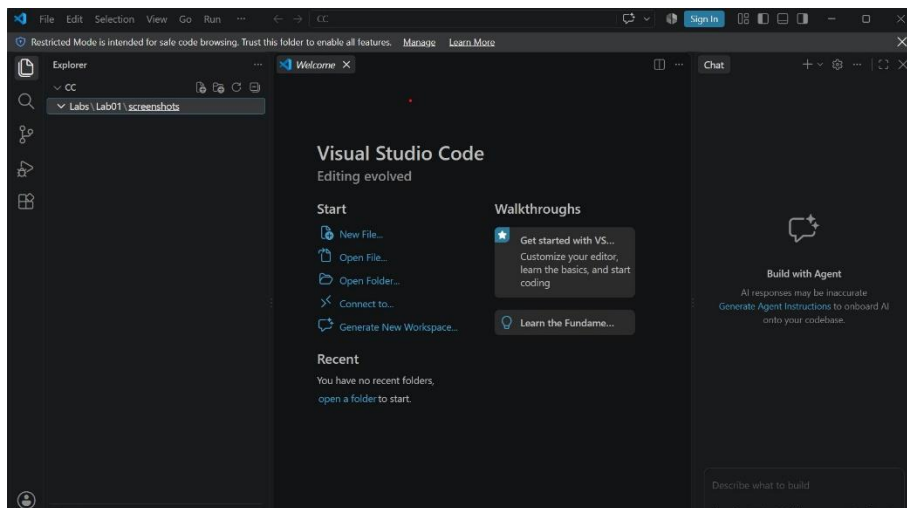
Section: BSE-V-A

Instructor: Prof. Waqas Saleem

Date: September 13, 2026

STUDENT INFORMATION

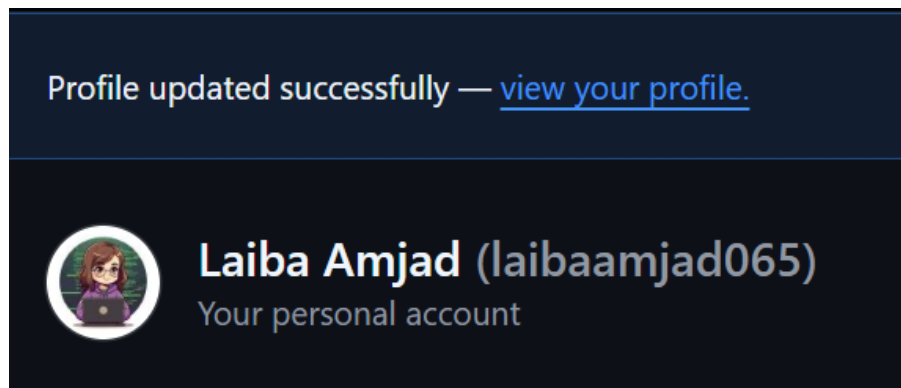
- **Student Name:** Laiba Amjad
- **Registration Number:** FA24B1-SE-021
- **Course Code:** CC-F26
- **Section:** BSE-V-A
- **GitHub Username:** laibaamjad065



Task 1: Create and Configure Your GitHub Account

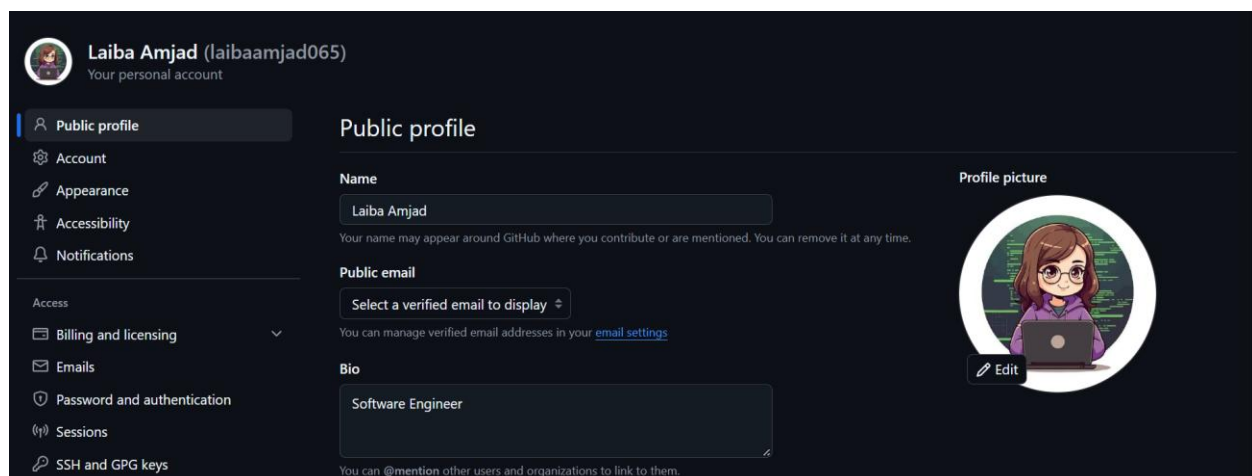
This task verifies the setup of the primary GitHub account used throughout the course.

Evidence 1.1: GitHub Profile



Caption: Screenshot showing public GitHub profile for user laibaamjad065.

Evidence 1.2: GitHub Actual Name Field

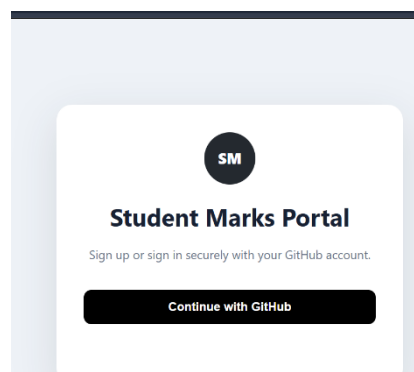


Caption: GitHub public settings page displaying complete actual name 'Laiba Amjad'.

Task 2: Register in the Student Marks Portal

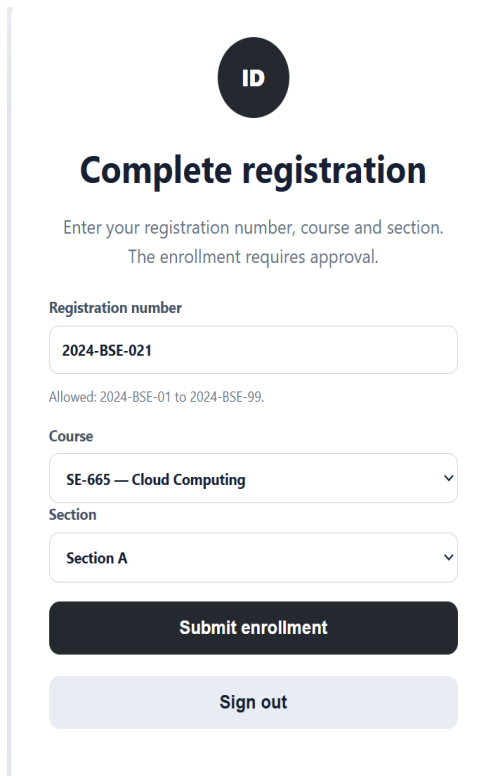
This task documents registration on the official Student Marks Portal using GitHub OAuth.

Evidence 2.1: Portal Login Page



Caption: Initial landing page of the Student Marks Portal showing GitHub Sign-In option.

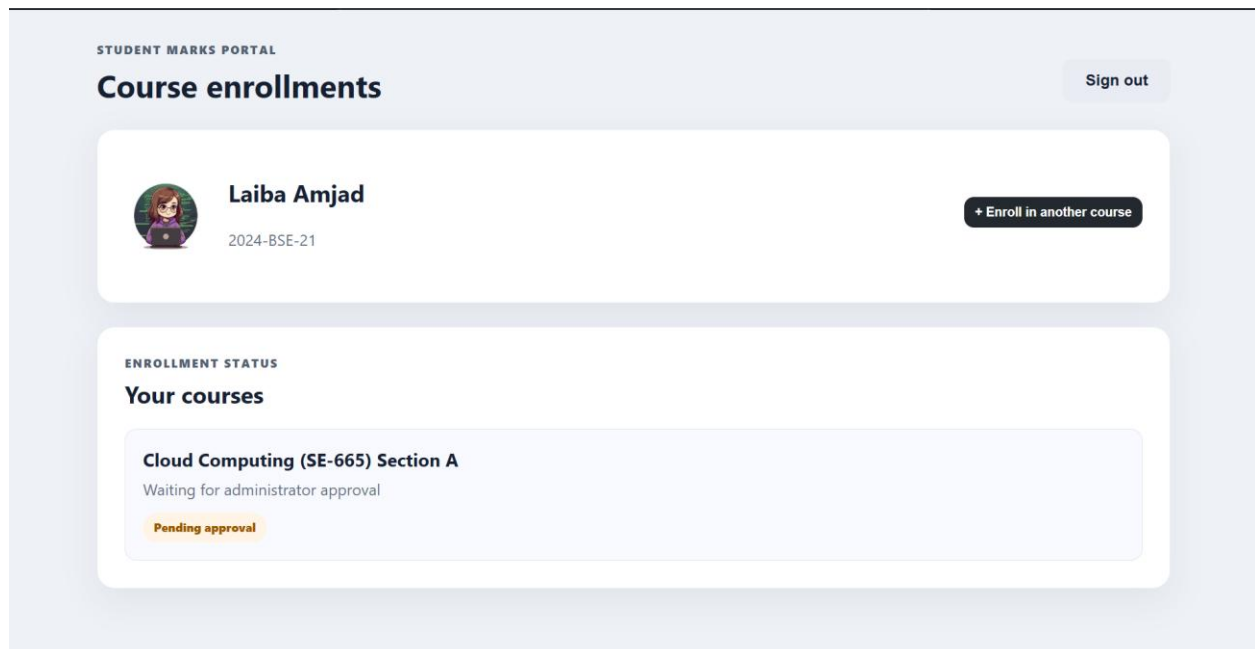
Evidence 2.2: Portal GitHub Profile Link



The screenshot shows a web interface for completing registration. At the top, there is a dark circular profile icon with the letters 'ID' in white. Below this is the heading 'Complete registration' in a bold, dark font. Underneath the heading, there is a prompt: 'Enter your registration number, course and section. The enrollment requires approval.' The form consists of three input fields: 'Registration number' with the value '2024-BSE-021', 'Course' with a dropdown menu showing 'SE-665 — Cloud Computing', and 'Section' with a dropdown menu showing 'Section A'. Below these fields are two buttons: a dark blue 'Submit enrollment' button and a light blue 'Sign out' button. A small text note below the registration number field states: 'Allowed: 2024-BSE-01 to 2024-BSE-99.'

Caption: Successful authentication into portal displaying linked GitHub account credentials.

Evidence 2.3: Enrollment Submission Confirmation

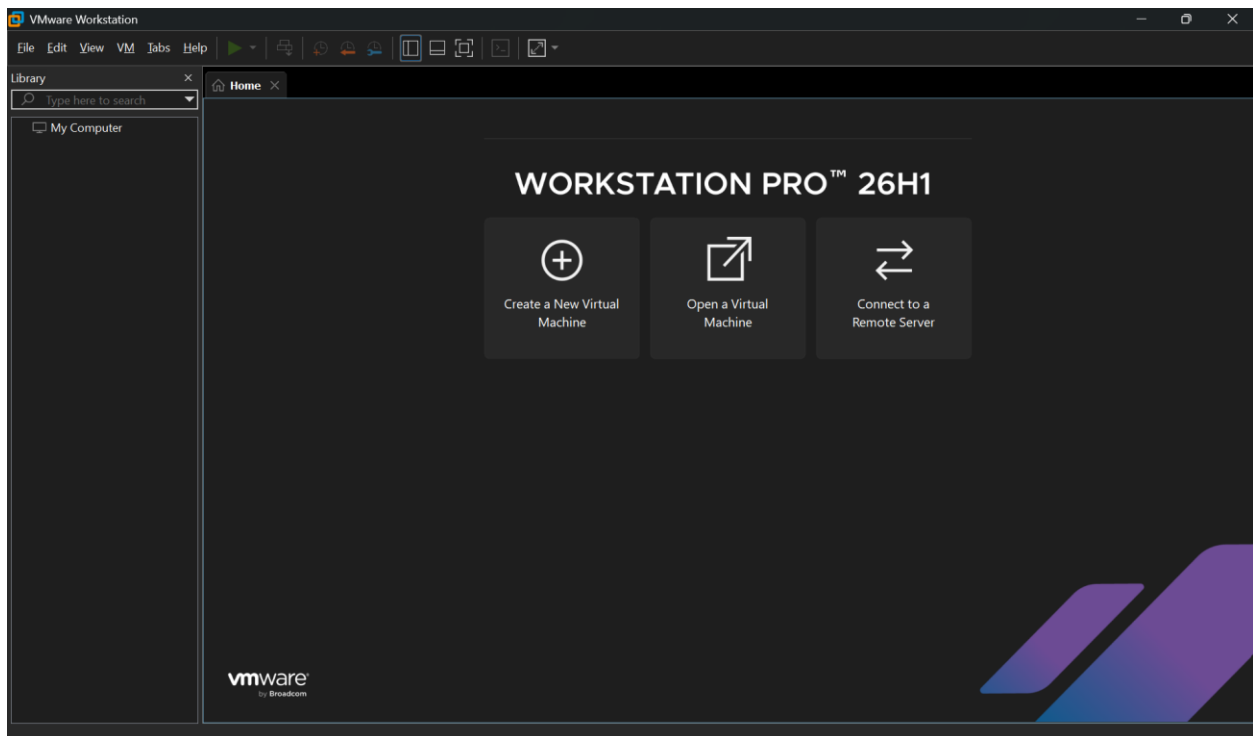


Caption: Portal dashboard confirming course enrollment submission for CC-F26, Section BSE-V-A.

Task 3: Download VMware Workstation Pro and Ubuntu Server

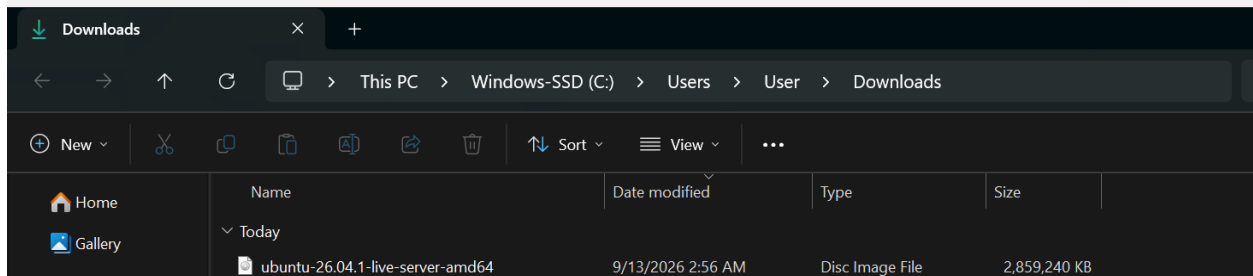
Verification of hypervisor installation, ISO download, and host machine disk capacity.

Evidence 3.1: VMware Workstation Pro Status



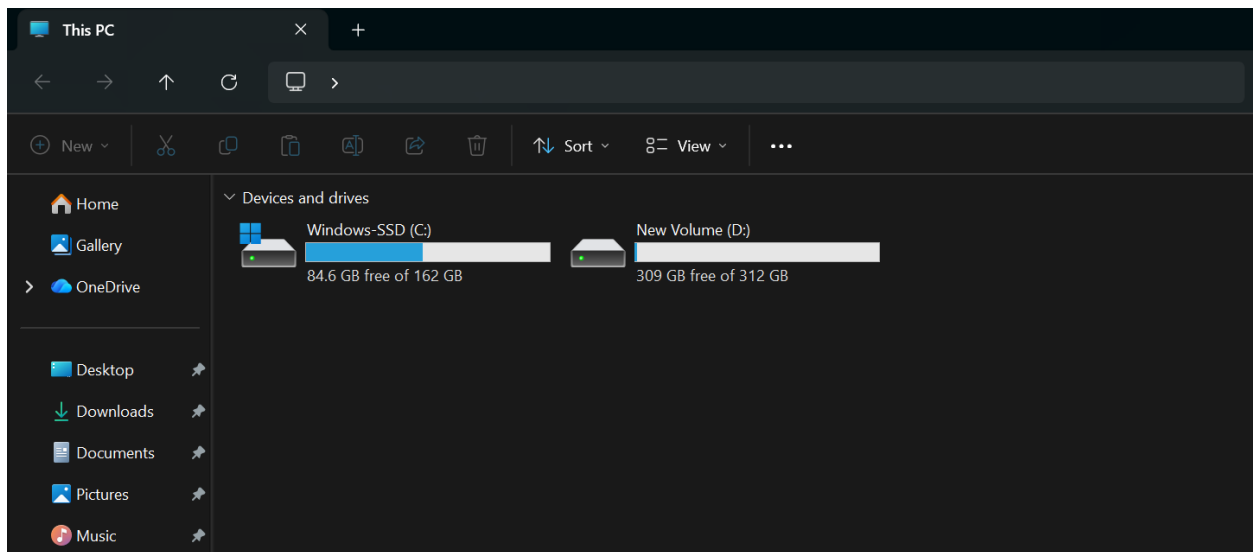
Caption: VMware Workstation Pro running on Windows host machine.

Evidence 3.2: Ubuntu Server ISO Verification



Caption: Downloaded Ubuntu Server 24.04 LTS ISO file in local storage directory.

Evidence 3.3: Storage Availability

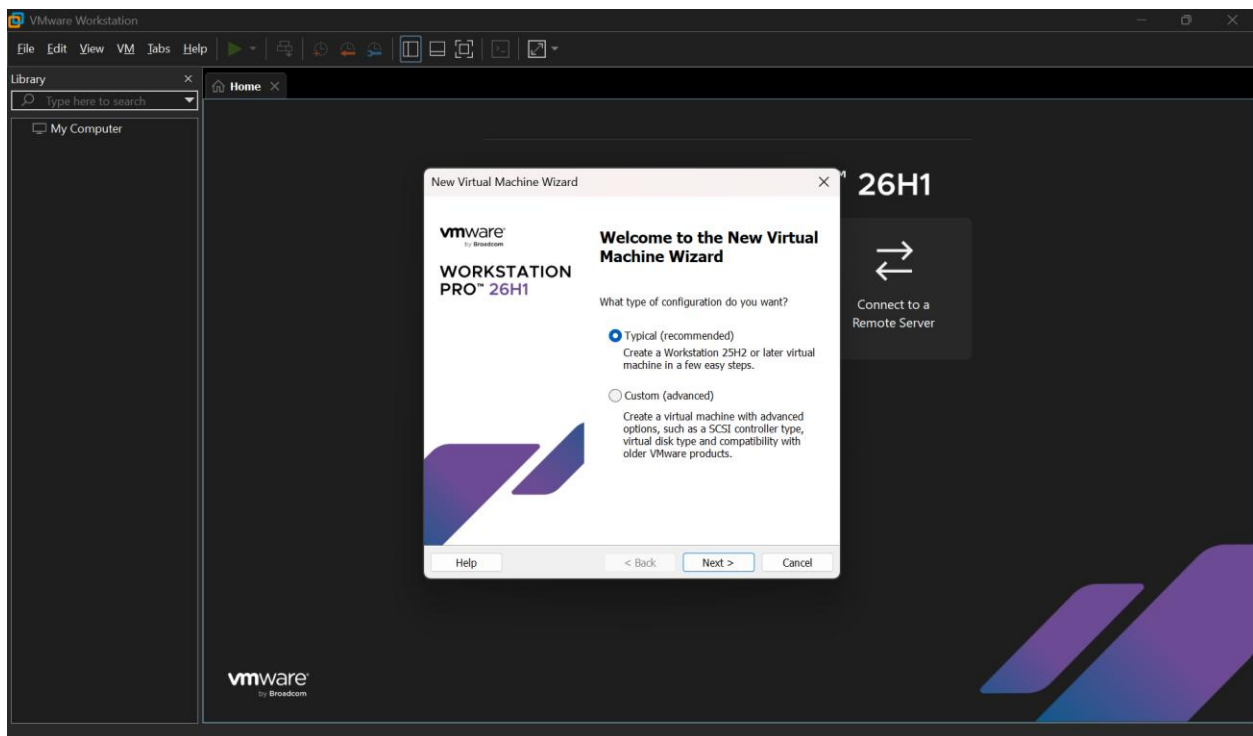


Caption: Host storage drive showing sufficient free disk space allocated for VM creation.

Task 4: Create the Ubuntu Server Virtual Machine

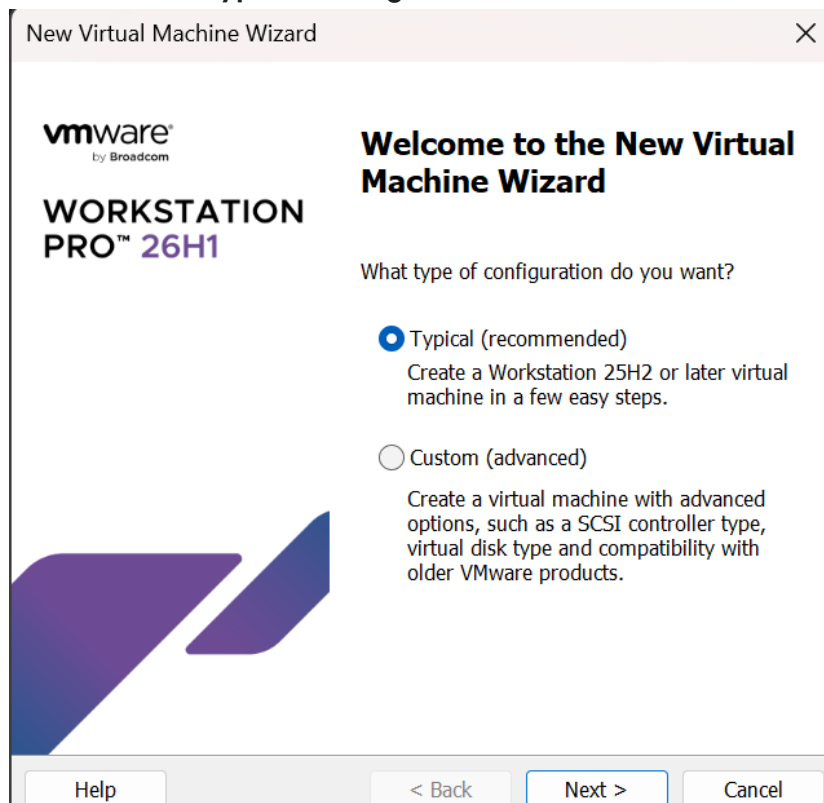
Step-by-step setup of the Virtual Machine hardware parameters within VMware.

Evidence 4.1: VM Creation Wizard



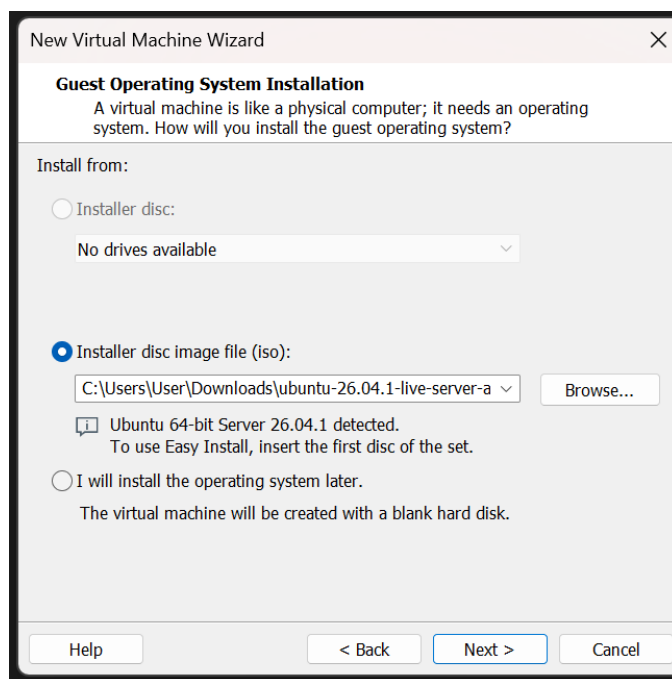
Caption: Opening screen of VMware Virtual Machine Creation Wizard.

Evidence 4.2: Typical Configuration Selection



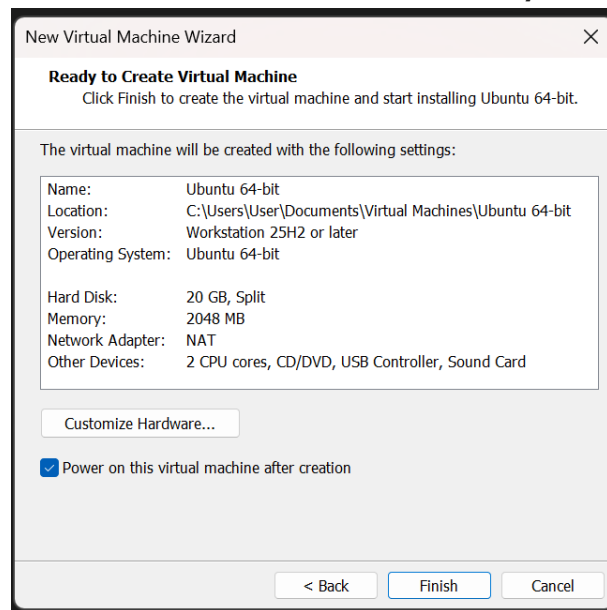
Caption: Selection of Typical (recommended) VM configuration option.

Evidence 4.3: ISO Mount Selection



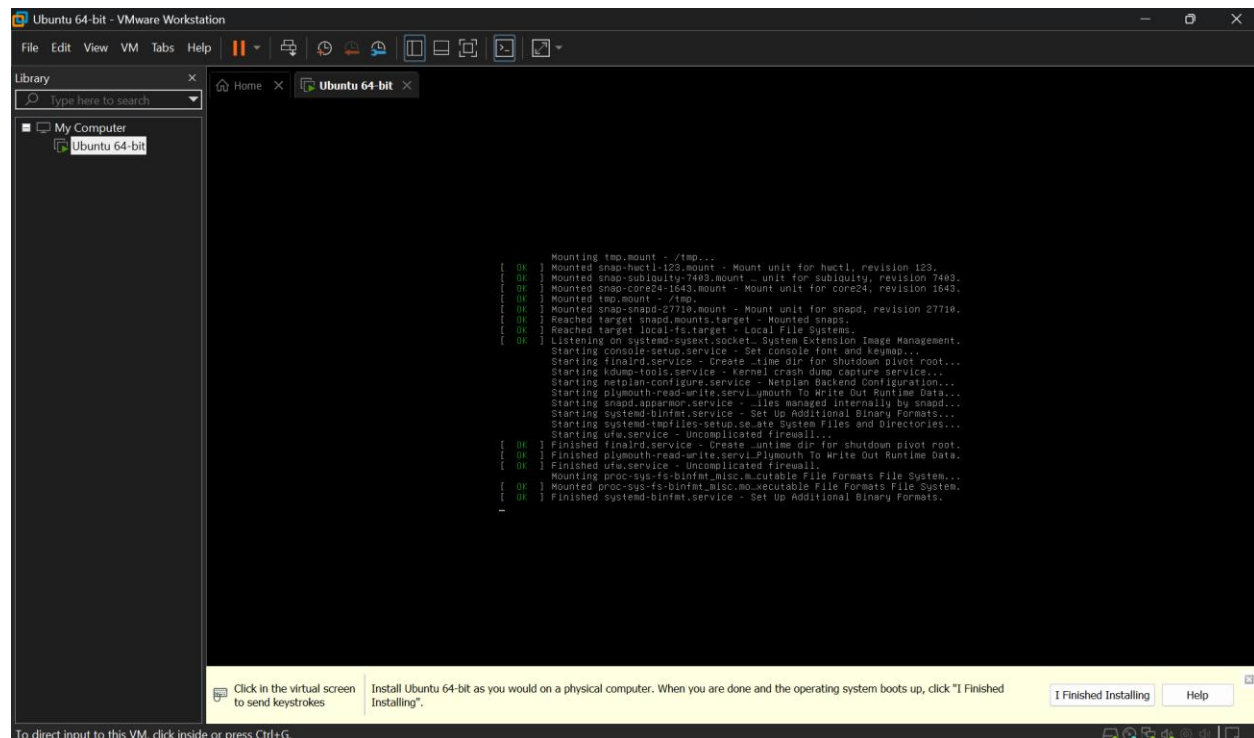
Caption: ISO path pointing to the downloaded Ubuntu Server image.

Evidence 4.4: VM Hardware Summary



Caption: Final configuration summary window showing CPU, RAM, and Disk allocations.

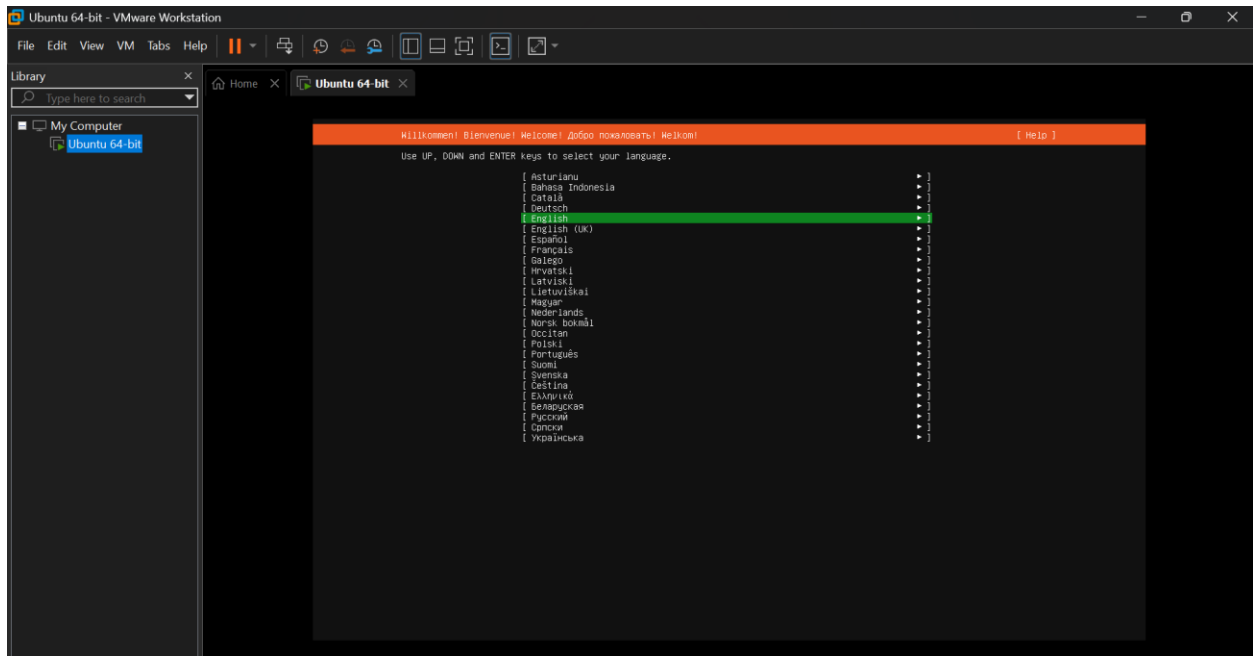
Evidence 4.5: Initial Boot Screen



Caption: Ubuntu Server GRUB boot loader screen upon initial VM power-on.

Task 5: Choose the Installation Language

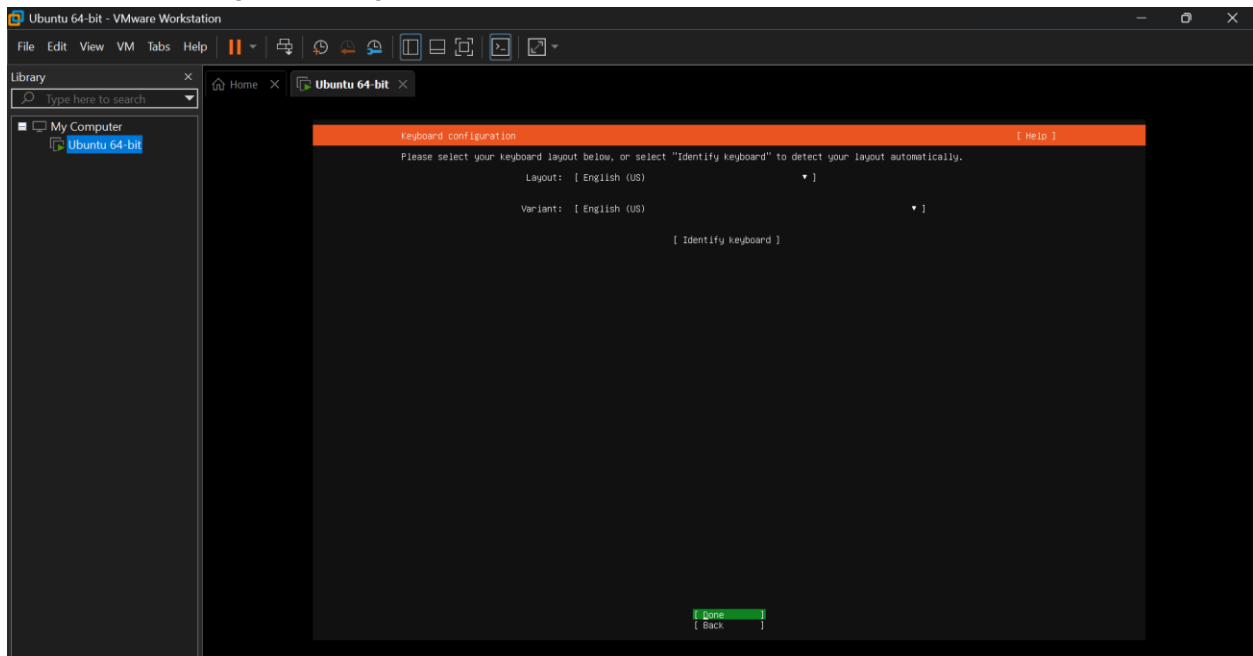
Evidence 5.1: Language Selection



Caption: Selecting English as the default installation language for Ubuntu Server.

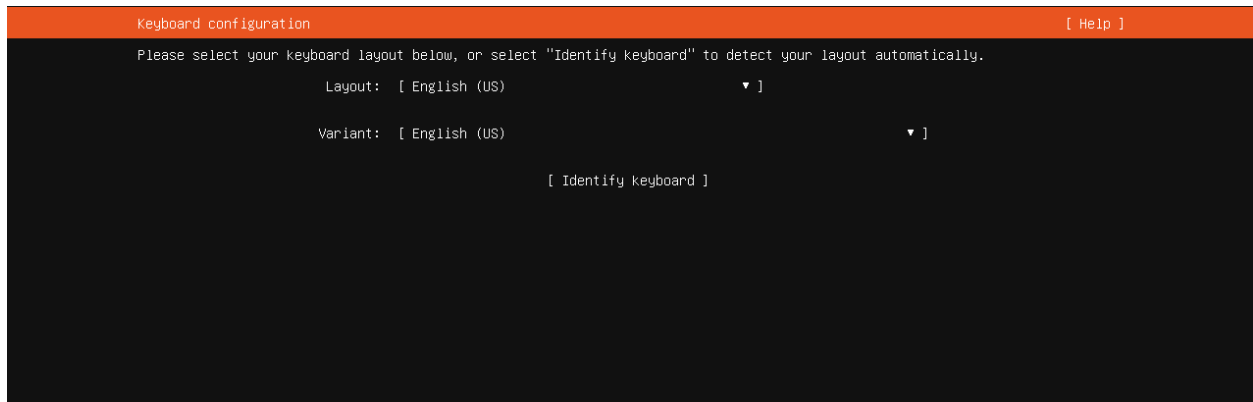
Task 6: Configure the Keyboard Layout

Evidence 6.1: Keyboard Layout



Caption: Setting English (US) as primary keyboard layout.

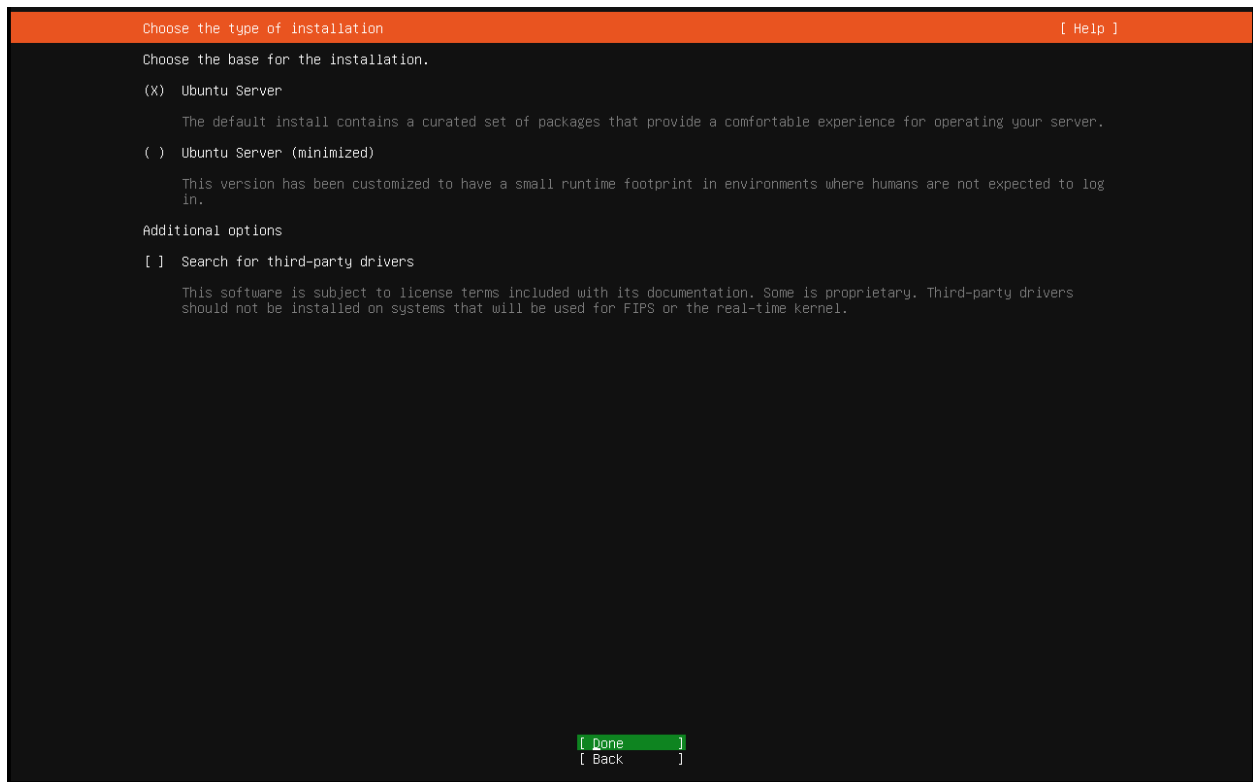
Evidence 6.2: Keyboard Variant



Caption: Confirming English (US) variant configuration.

Task 7: Choose the Ubuntu Server Installation Type

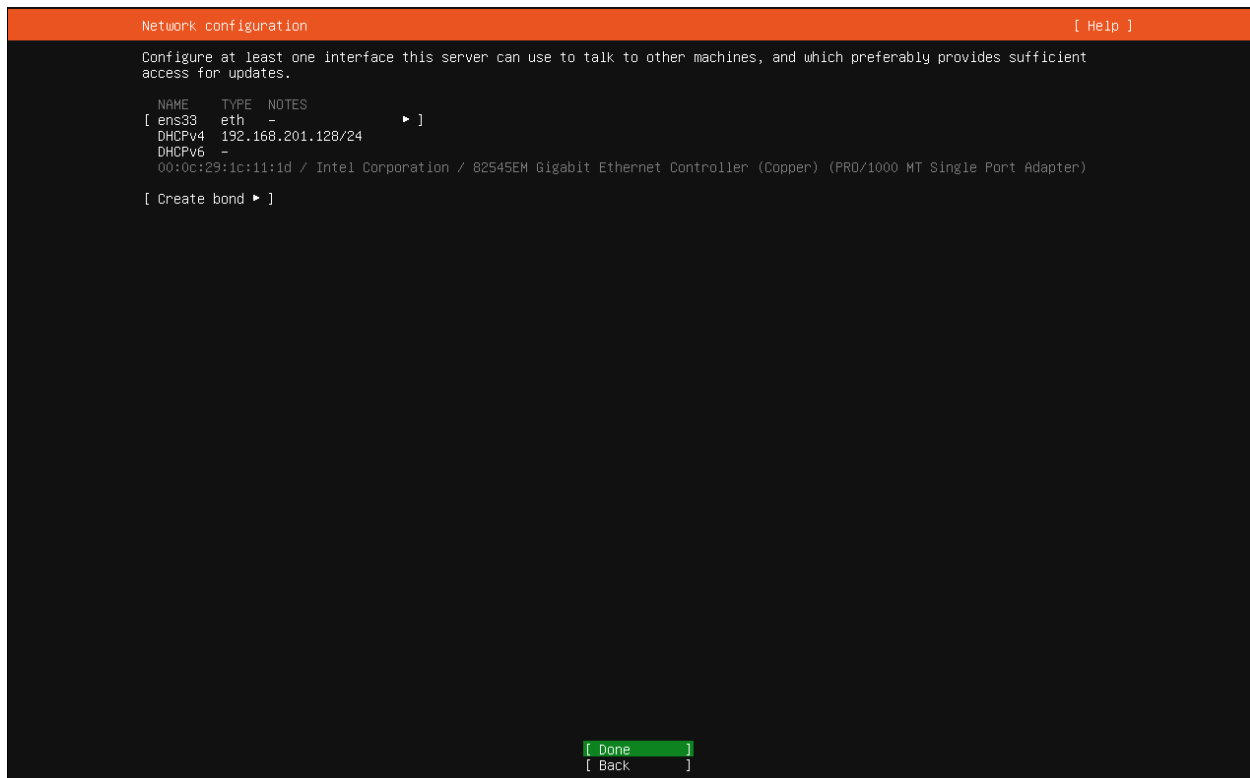
Evidence 7.1: Installation Type



Caption: Selecting standard Ubuntu Server installation profile.

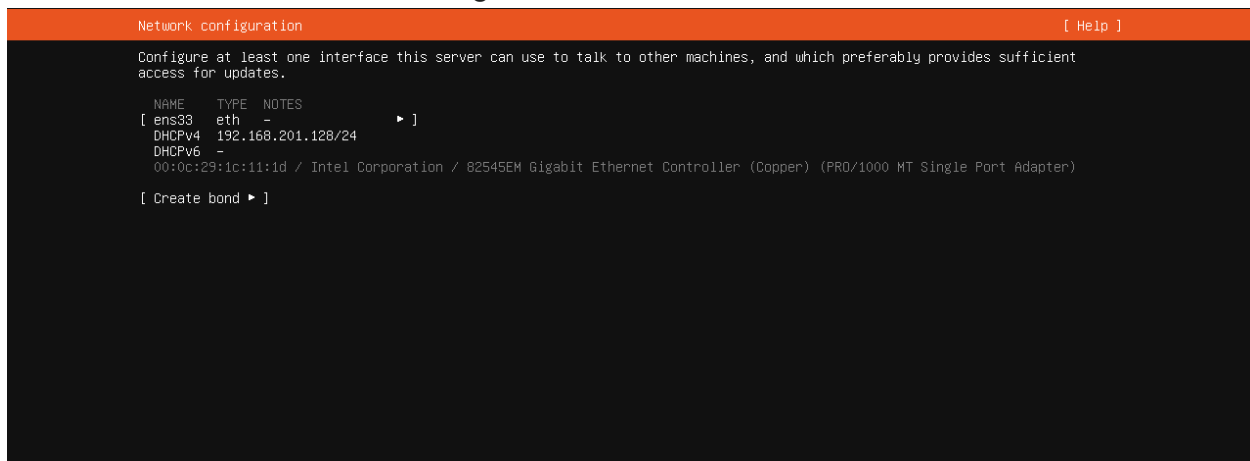
Task 8: Configure the Network

Evidence 8.1: Network Interface Detection



Caption: Installer detecting primary network interface ens33.

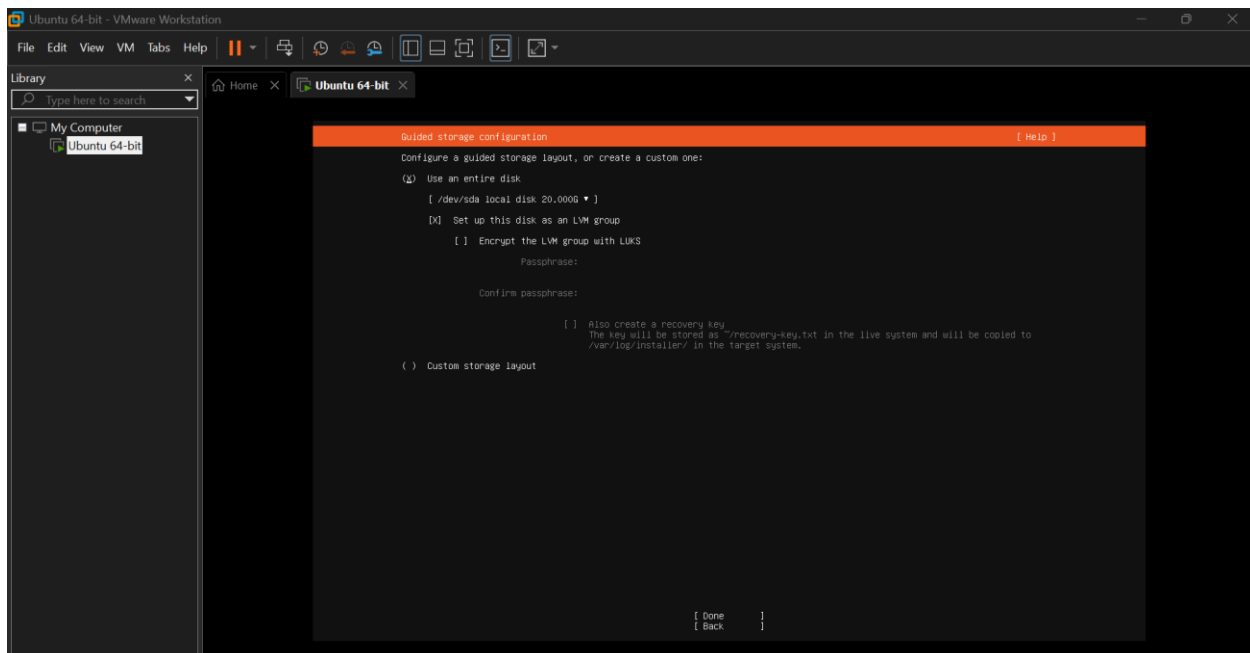
Evidence 8.2: DHCP Address Assignment



Caption: Successful automatic IPv4 address assignment via DHCP.

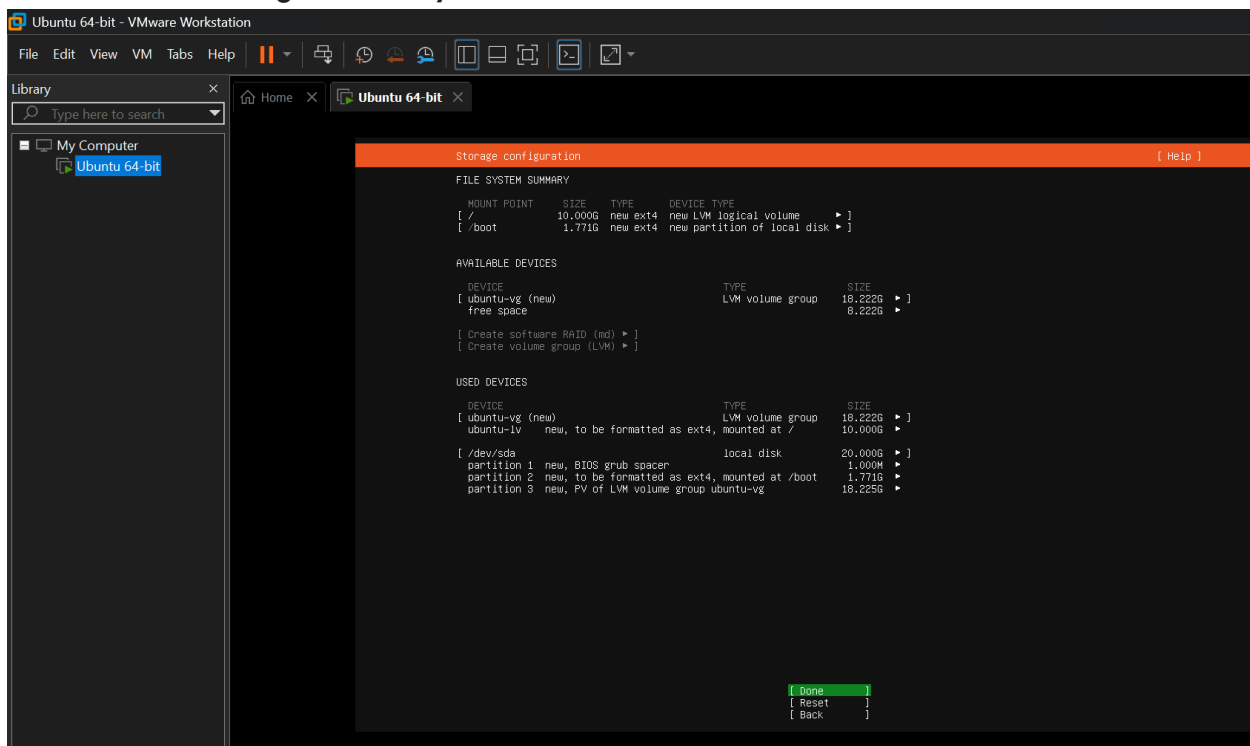
Task 9: Configure Storage

Evidence 9.1: Guided Storage Setup



Caption: Choosing guided storage layout utilizing entire virtual disk.

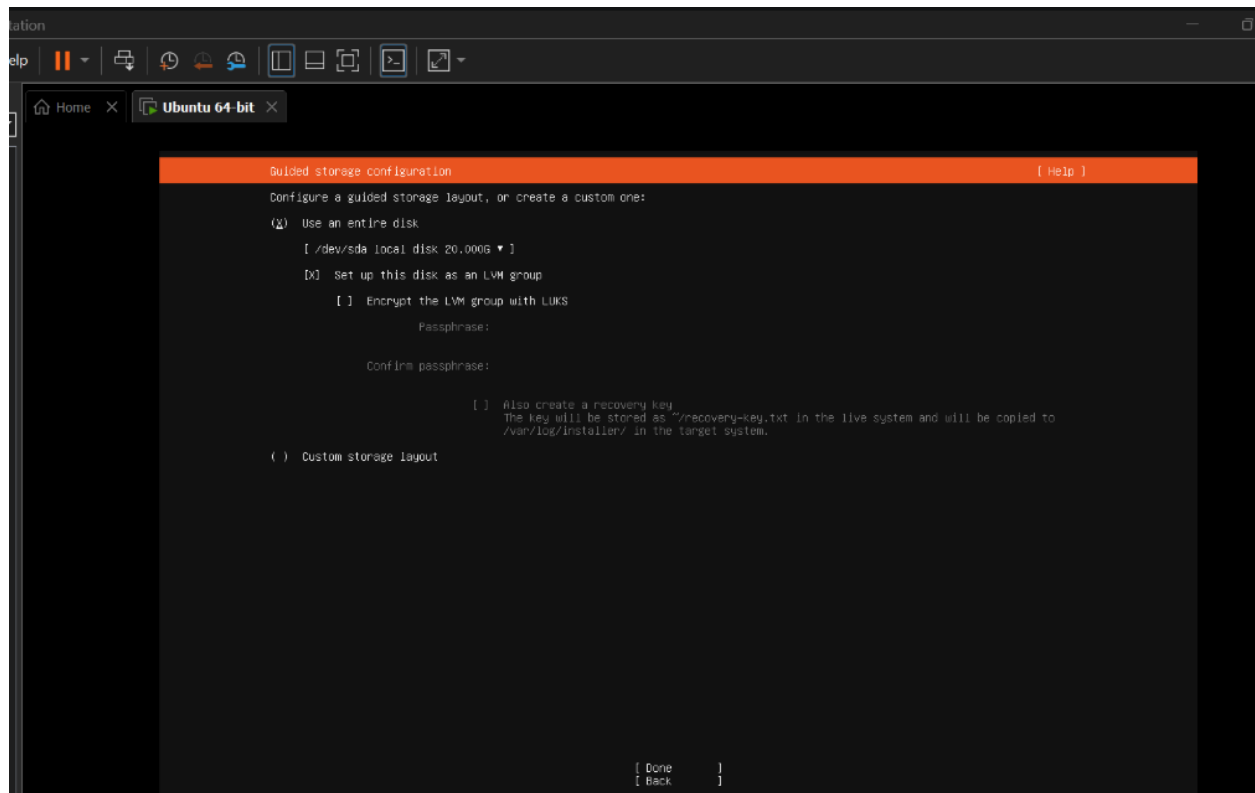
Evidence 9.2: Storage Summary



Caption: Reviewing overall storage layout before disk allocation.

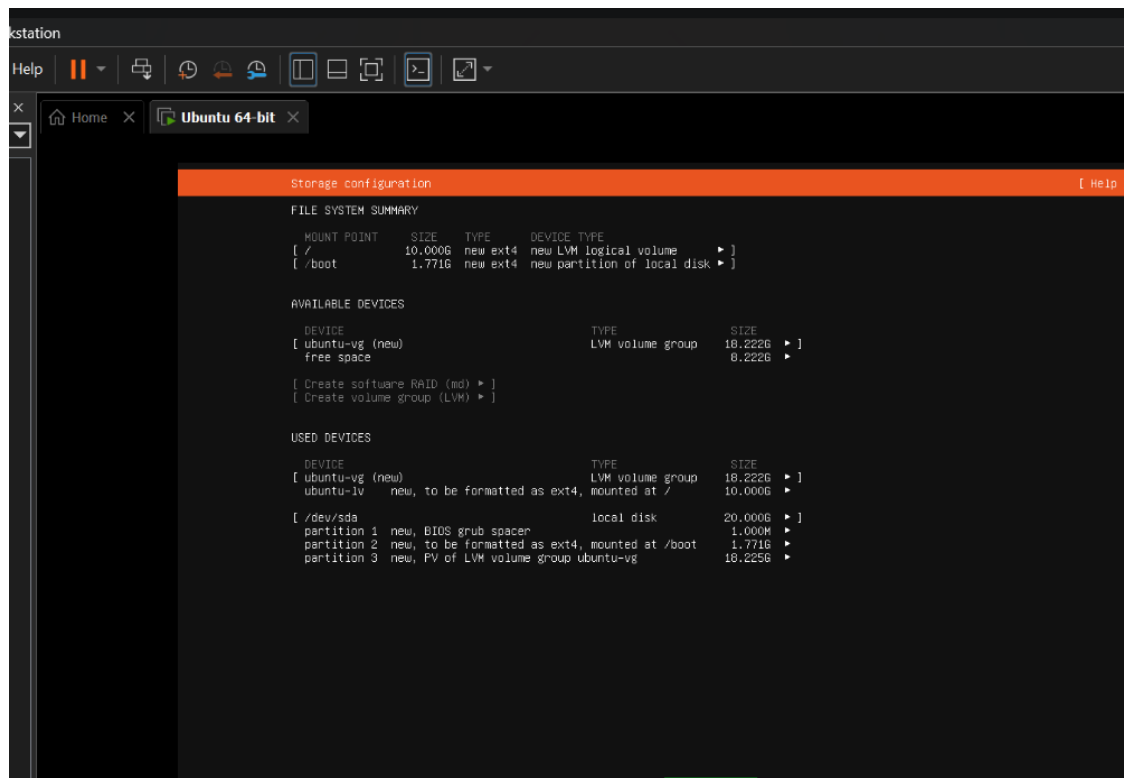
Task 10: Select the Installation Disk

Evidence 10.1: Disk Selection



Caption: Selecting target VMware Virtual Disk for installation.

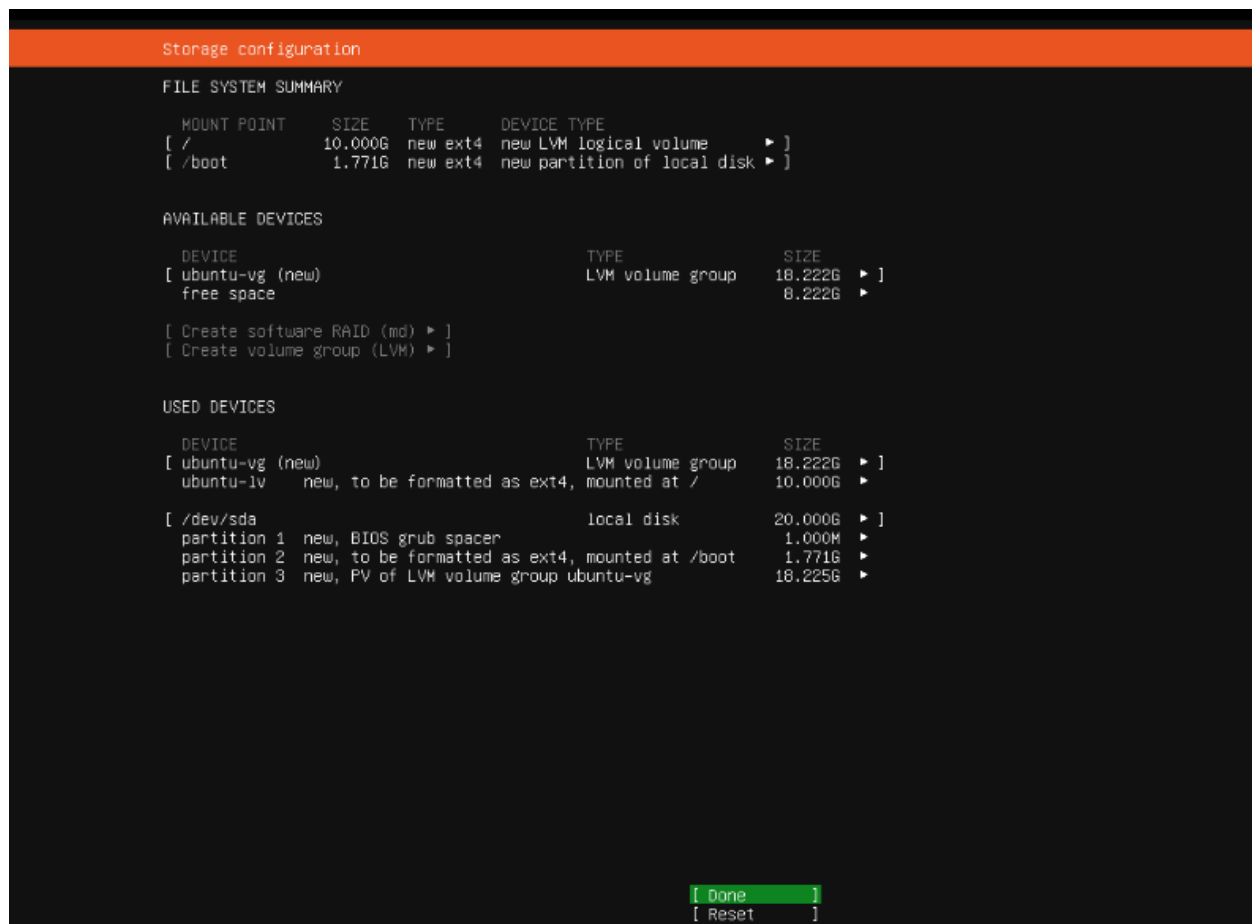
Evidence 10.2: Disk Size Verification



Caption: Verifying virtual disk size matches allocated capacity (20GB+).

Task 11: Review the Partitions

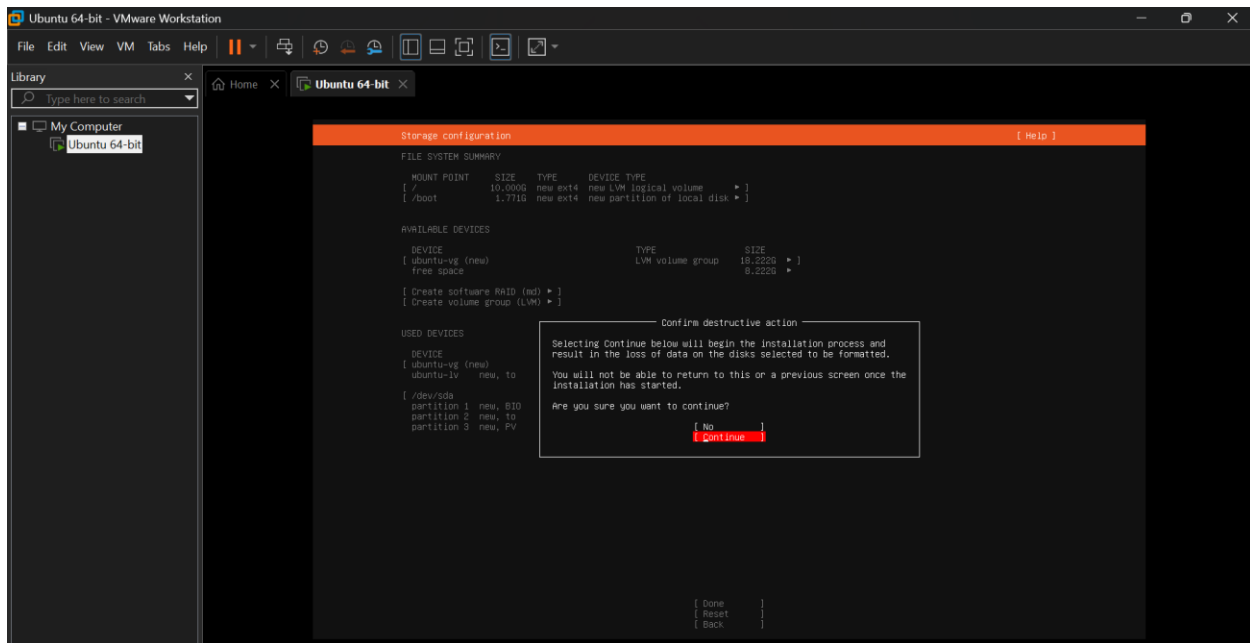
Evidence 11.1: Partition Scheme Layout



Caption: Detailed view of root (/) and boot partition assignments.

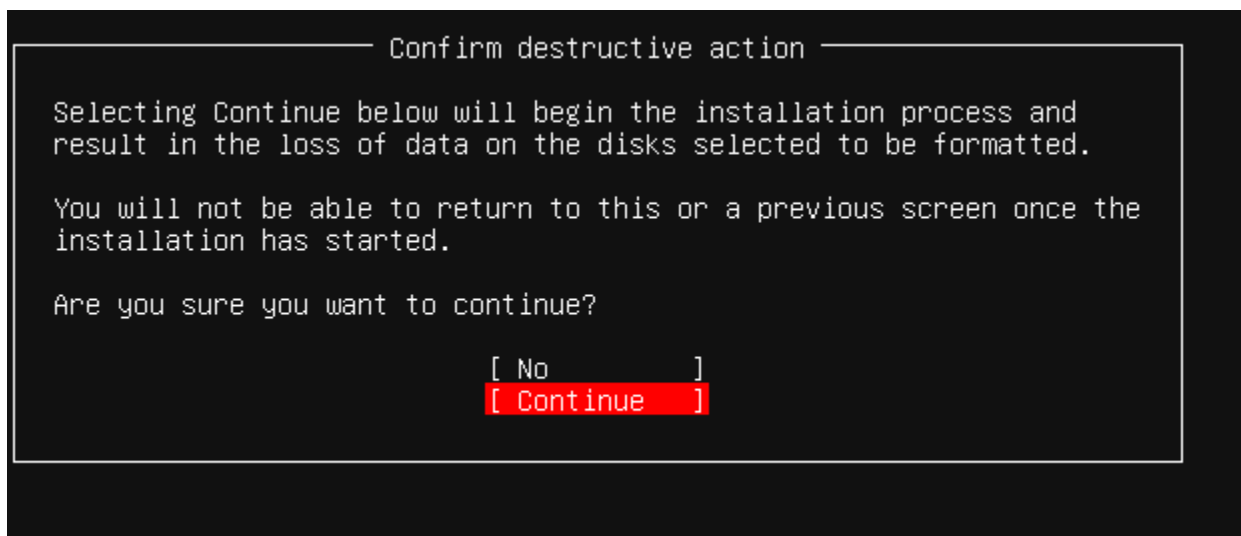
Task 12: Confirm the Storage Changes

Evidence 12.1: Storage Destructive Action Warning



Caption: Installer confirmation dialog warning about writing changes to disk.

Evidence 12.2: Execution Confirmation

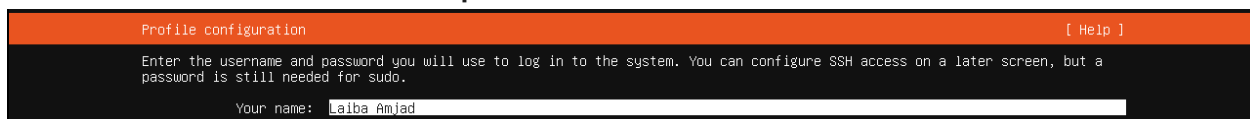


Caption: Selecting 'Continue' to initiate disk formatting and partitioning.

Task 13: Configure the Ubuntu Profile

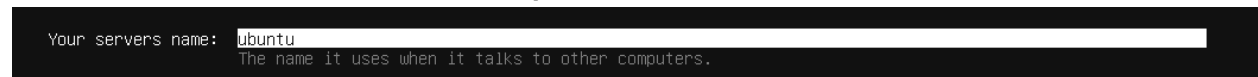
Setting up system administrative account identity.

Evidence 13.1: Actual Name Setup



Caption: Profile configuration screen entering 'Laiba Amjad' in Your Name field.

Evidence 13.2: Server Hostname Setup



Your servers name:

The name it uses when it talks to other computers.

Caption: Setting system hostname as ubuntu.

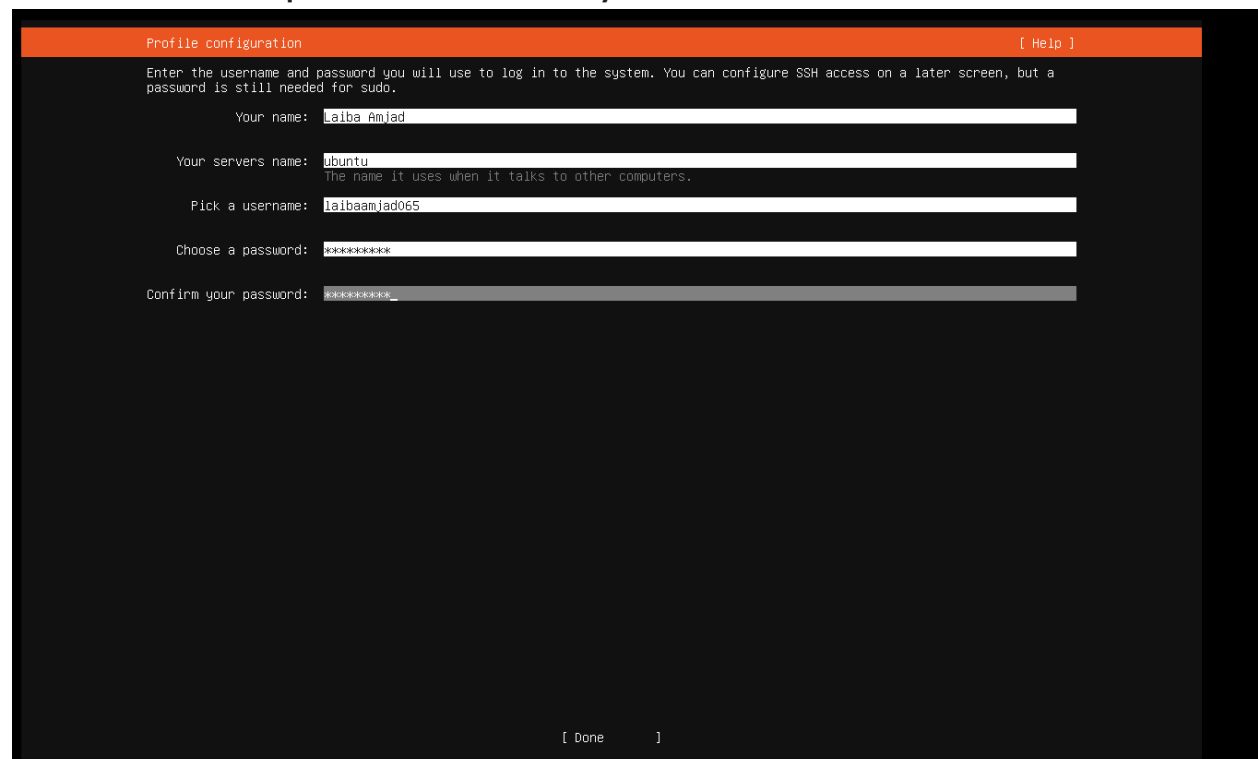
Evidence 13.3: System Username Setup



Pick a username:

Caption: Setting system username to match GitHub username laibaamjad065.

Evidence 13.4: Completed Profile Summary



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

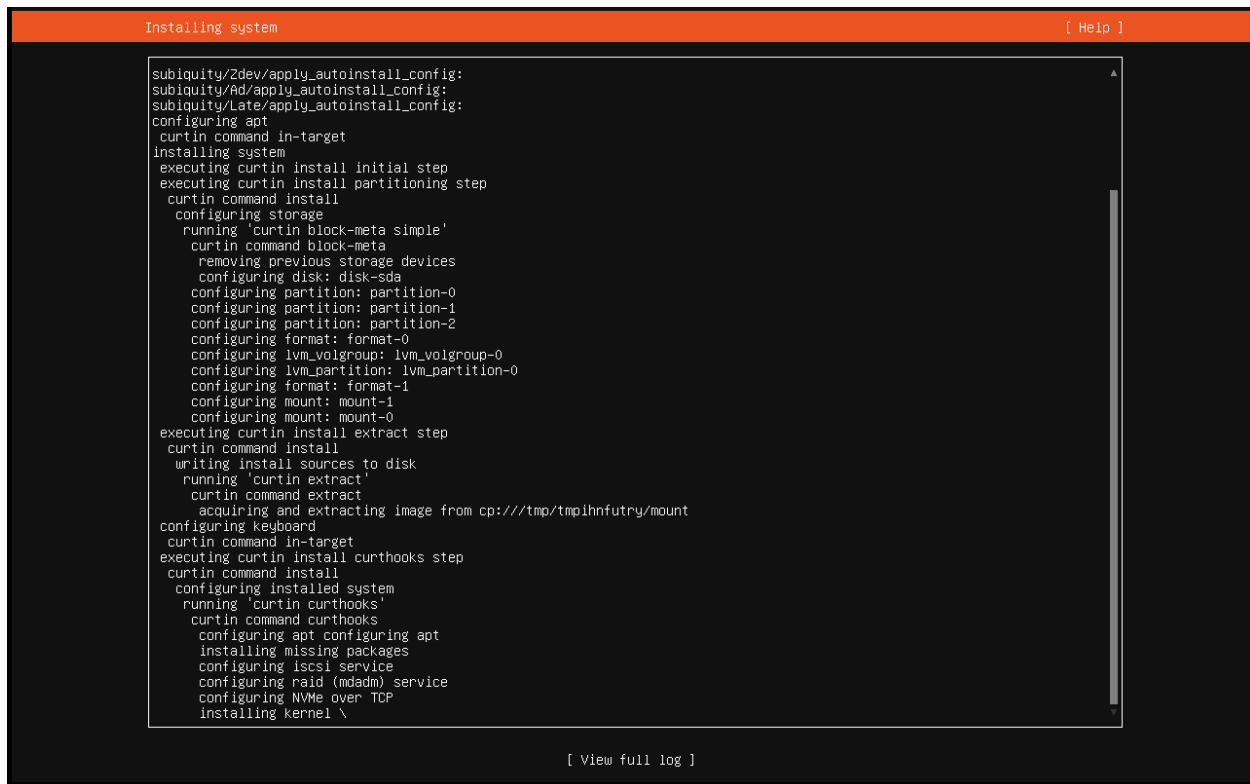
Confirm your password:

[Done]

Caption: Complete user profile setup screen prior to system installation.

Task 14: Complete the Software Installation

Evidence 14.1: Software Extraction Progress



```
Installing system [ Help ]

subiquity/2dev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
  configuring storage
    running 'curtin block-meta simple'
    curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
      configuring partition: partition-0
      configuring partition: partition-1
      configuring partition: partition-2
      configuring format: format-0
      configuring lvm_voigroup: lvm_voigroup-0
      configuring lvm_partition: lvm_partition-0
      configuring format: format-1
      configuring mount: mount-1
      configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp1hmfutry/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel \
```

[View full log]

Caption: Core system packages unpacking and installation progress bar.

Task 15: Reboot and Verify the Ubuntu Identity

System installation finalization, rebooting, and initial shell authentication.

Evidence 15.1: Installation Complete Screen

```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tpmihnfutry/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Caption: Installer screen confirming successful software setup and prompt to reboot.

Evidence 15.2: TTY Login Prompt

```
[ 22.922904] cloud-init[1152]: | . + +o o|
[ 22.929710] cloud-init[1152]: | o . .++o|
[ 22.933543] cloud-init[1152]: | . o.=+o|
[ 22.935899] cloud-init[1152]: | S ..B=+o|
[ 22.937230] cloud-init[1152]: | + o*+=+|
[ 22.937527] cloud-init[1152]: | . +.o0=+|
[ 22.937769] cloud-init[1152]: | o E=+o|
[ 22.937973] cloud-init[1152]: | . .o=o|
[ 22.938842] cloud-init[1152]: +----[SHA256]-----+
[ 22.940450] cloud-init[1152]: Generating public/private ed25519 key pair.
[ 22.943197] cloud-init[1152]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 22.947744] cloud-init[1152]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 22.949486] cloud-init[1152]: The key fingerprint is:
[ 22.950718] cloud-init[1152]: SHA256:kot15NcSwi7a3aasA7ZpA23GnZx20T5JE5+uynM9WBI root@ubuntu
[ 22.952126] cloud-init[1152]: The key's randomart image is:
[ 22.953310] cloud-init[1152]: +--[ED25519 256]--+
[ 22.954634] cloud-init[1152]: | .o |
[ 22.956244] cloud-init[1152]: | .*. |
[ 22.960383] cloud-init[1152]: | o + B+++ |
[ 22.962758] cloud-init[1152]: | . = E+++ o |
[ 22.966515] cloud-init[1152]: | o ..B.S . |
[ 22.969260] cloud-init[1152]: | +o.*.* . |
[ 22.971533] cloud-init[1152]: | ..=o++ o |
[ 22.973529] cloud-init[1152]: | +.+++o |
[ 22.975825] cloud-init[1152]: | . ==+. |
[ 22.978560] cloud-init[1152]: +----[SHA256]-----+

ubuntu login:
```

Caption: System startup completing at the ubuntu login: prompt.

Evidence 15.3: Successful Shell Login

```
ubuntu login: laibaamjad065
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 12:23:26 PM UTC 2026

System load:  0.17               Processes:            249
Usage of /:   50.5% of 9.75GB    Users logged in:     0
Memory usage: 22%               IPv4 address for ens33: 192.168.201.129
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibaamjad065@ubuntu:~$ _
```

Caption: Welcome banner (MOTD) displayed after successful password authentication.

Evidence 15.4: Terminal Identity Verification

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibaamjad065@ubuntu:~$
```

Caption: Active Bash terminal prompt displaying identity laibaamjad065@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

Evidence 16.1: Execution of ip addr Command

```

laibaamjad065@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:1c:11:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c291c111d
    inet 192.168.201.129/24 metric 100 brd 192.168.201.255 scope global dynamic ens33
        valid_lft 1412sec preferred_lft 1412sec
    inet6 fe80::20c:29ff:fe1c:111d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
laibaamjad065@ubuntu:~$

```

Caption: Execution of ip addr command in local server terminal.

Evidence 16.2: Active IPv4 Address Identification

```

laibaamjad065@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:1c:11:1d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c291c111d
    inet 192.168.201.129/24 metric 100 brd 192.168.201.255 scope global dynamic ens33
        valid_lft 1412sec preferred_lft 1412sec
    inet6 fe80::20c:29ff:fe1c:111d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
laibaamjad065@ubuntu:~$ _

```

Caption: Terminal output displaying interface ens33 IPv4 address 192.168.201.129 alongside terminal prompt.

Task 17: Connect to Ubuntu Server from Windows

Remote administration setup using SSH protocol from Windows Host Command Prompt.

Evidence 17.1: SSH Command Execution

```

C:\Users\User>ssh laibaamjad065@192.168.201.129
The authenticity of host '192.168.201.129 (192.168.201.129)' can't be established.
ED25519 key fingerprint is SHA256:kotl5NcSwi7a3aasA7ZpAZ3GnZxZ0T5JE5+uyrM9WBI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |

```

Caption: Windows CMD executing command ssh laibaamjad065@192.168.201.129.

Evidence 17.2: Key Fingerprint Acceptance

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.201.129' (ED25519) to the list of known hosts.
laibaamjad065@192.168.201.129's password: |
```

Caption: Host key authenticity prompt accepted by typing 'yes' to store ED25519 fingerprint.

Evidence 17.3: Remote SSH Session Established

```
laibaamjad065@ubuntu: ~
C:\Users\User>ssh laibaamjad065@192.168.201.129
laibaamjad065@192.168.201.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 12:37:04 PM UTC 2026

System load:  0.0                Processes:            223
Usage of /:   50.8% of 9.75GB     Users logged in:     0
Memory usage: 25%                IPv4 address for ens33: 192.168.201.129
Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

3 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

laibaamjad065@ubuntu:~$ |
```

Caption: Remote session banner display following password verification.

Evidence 17.4: Remote SSH Shell Identity

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

laibaamjad065@ubuntu:~$ |
```

Caption: Remote terminal prompt verification displaying laibaamjad065@ubuntu:~\$ over SSH.

